Work and the Work-Energy Relation

Synopsis: We left the skater on her ramp with a problem. With no friction she can never get off it — down one side, up the other, back again, forever. It is friction that lets her stop. So friction is not just something that spoils the arithmetic: it is doing something real, and something we ought to be able to measure.

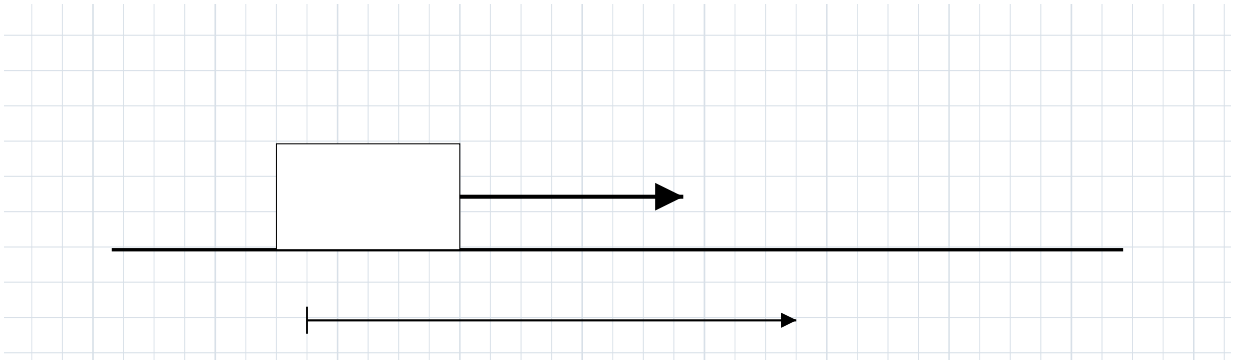
This Note builds the tool. We will work out what it means for a force to change an object's energy, and then take that tool first to a cart crossing a rough patch and then to a skier on a hill, and find out how much energy friction took in each case — using nothing but the speeds and the height.

Performance Expectations: Calculate the work done by a force acting through a distance. Explain how the work done on an object changes its kinetic energy. Separate the work done by gravity from the work done by everything else, and use the difference to show that friction acted and to find how much energy it took.

For PHY 151 — with vectors: Resolve a force into the part along the displacement and the part across it, and evaluate the work as a dot product.

Instructions and Questions

1) What work is.
   Energy does not change on its own. Something has to push or pull an object through a distance, and that is what we call work. Work, W, is measured in joules (J) — the same unit as energy, which is no coincidence: work is energy being handed over. When the force points the way the object is moving, W = Fd, where F is the force in newtons (N) and d is the distance moved in metres (m).



   i) A 20 N force pushes a crate 100 m along a level floor, in the direction of the push. How much work does the force do?
   ii) Work and energy share a unit. Say in one sentence why that is not an accident.

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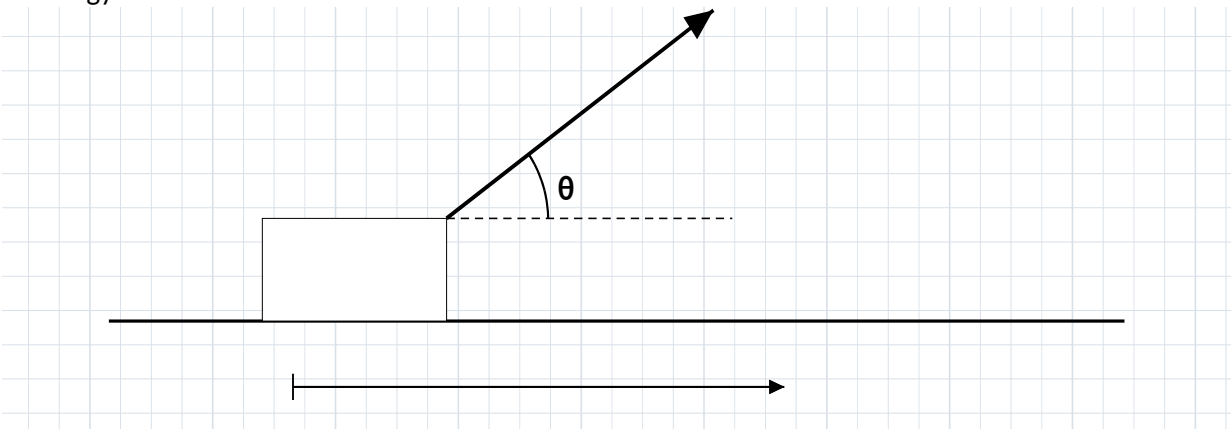
2) A force only does work if the object moves along it.
A force does not have to point the way the object is going, and when it does not, it does less work — or none at all. Push the crate forwards and you add energy to it. Friction pushes backwards against the motion and takes energy away. The floor pushes straight up on the crate while the crate slides sideways, and that force does no work whatsoever, however hard it pushes: the crate never moves along it.

with the motion	against the motion	square to the motion

i) A force pushes a crate forwards as it slides forwards. Does it add energy or take it away?
ii) Friction acts backwards on that crate. Does it add energy or take it away?
iii) The floor pushes up on the crate with a force of 200 N while the crate slides 5 m sideways. How much work does the floor do? Explain your answer.

Putting a number on the in-between case.
The three cases above are the ends and the middle. For a force at any angle in between, only the part of it lying along the displacement does work; the part across the motion does none. Force and displacement are both vectors, so written properly the work is $W = F \cdot D = |F||D| \cos\theta$, where θ is the angle between them. The dot product turns two vectors into a single number — work is a scalar. It carries a sign, but no direction.

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i) The same 20 N force now acts at 30° above the horizontal while the crate still slides 100 m along the floor. How much work does it do?
ii) Check your third answer above against the formula. What is cos 90°, and what does that make the work?
iii) Write F = aî + bĵ + ck̂ and D = dî + fĵ + gk̂, then write out F · D. Is your answer a vector or a number?

3) Work changes kinetic energy.
Do work on an object and its kinetic energy changes by exactly that amount. That is the work-energy theorem: ΔKE = W. The W here is the total work done on the object by every force acting on it, not by one force you happen to be interested in.

i) Write the kinetic energy at each instant, then write ΔKE in terms of m, v_i and v_f.
ii) State the work-energy theorem in a sentence a classmate would understand.

4) Two kinds of work, and a system that leaks.

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Think of the skater and her ramp as one closed system. Left alone, its total energy E = KE + PE stays constant: that is what conservation means, and it is why she never gets off. Work is how energy gets into or out of a system from outside, so once friction acts the system leaks, and $E_i + W_{NC} = E_f$. That is the work-energy relation.

The leak happens because forces do not all behave the same way. Gravity gives back exactly what it takes: lift an object and gravity takes energy from it, drop it and gravity hands the same energy straight back. Friction never gives anything back — it only ever takes. So we split the total work into those two kinds, $W = W_C + W_{NC}$, and the theorem becomes $\Delta KE = W_C + W_{NC}$. The work gravity does is precisely the potential energy the object loses: $W_C = -\Delta PE$.

i) Friction always dissipates. Gravity can speed an object up or slow it down. Explain why only one of the two can ever give the energy back.
ii) Say what W_C collects and what W_{NC} collects.
iii) With no friction the skater never escapes the ramp. Say what friction lets her do, in terms of the system's total energy.

No friction	With friction

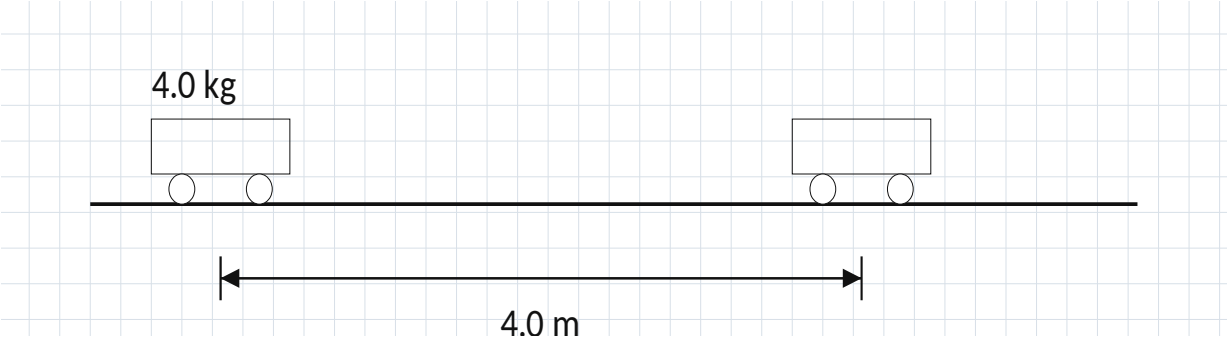
5) Finding friction without measuring it.

Here is the move this Note has been building to. Over any trip there are two things we can usually get at: how fast the object is going at each end, which gives us ΔKE, and how far it rose or fell, which gives us W_C. If gravity were the only thing doing work, those two would match. So subtract them. If $\Delta KE - W_C$ comes out zero, nothing else did any work. If it does not come out zero, something else did — and the leftover is exactly how much energy it took.

First, a trip with no height change at all.

A 4.0 kg cart is rolling along a level bench at 6.0 m/s. It crosses a rough patch 4.0 m long and comes off it doing 2.0 m/s.

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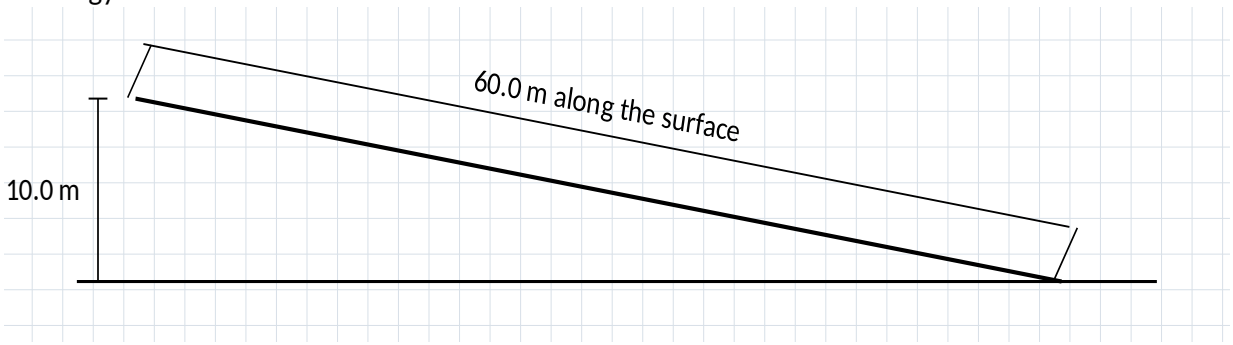


i) The bench is level. What does that make W_C, and why?
ii) Calculate ΔKE for the crossing.
iii) Work out ΔKE − W_C. With one of the two terms gone, what is left?
iv) The patch is 4.0 m long. How big was the friction force?

Now let gravity back in.

A 50 kg skier starts from rest at the top of a gentle hill and rolls down. The slope is 60.0 m long and drops 10.0 m. She reaches the bottom travelling at 10.0 m/s. Take g = 9.8 m/s². This time both terms are alive, and the method is the one you just used.

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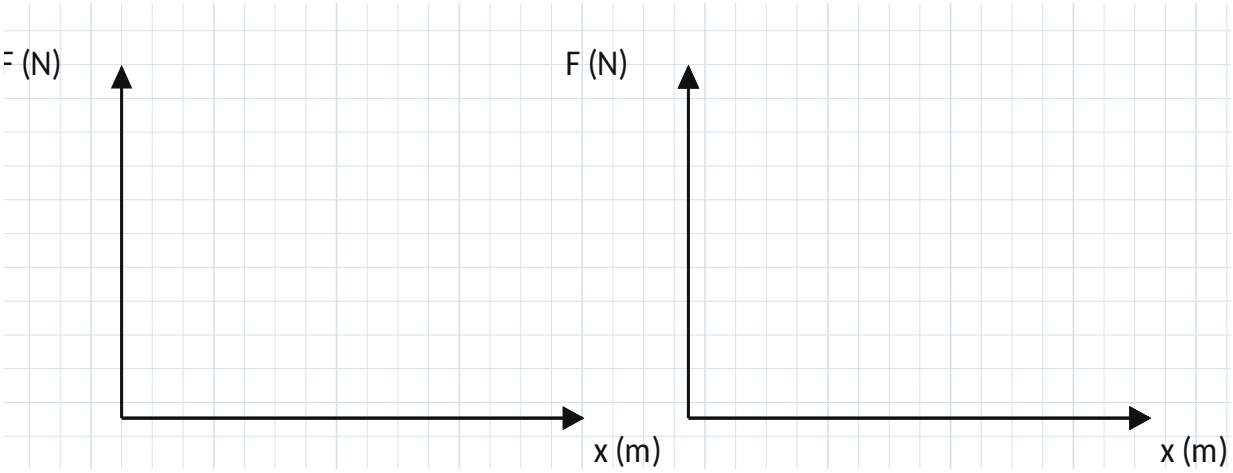
i) On the figure, mark where she starts and where she finishes, her speed at each end, and the drop.
ii) How fast would she have been going at the bottom if gravity were the only force doing work on her?
iii) She is actually going 10.0 m/s. Calculate ΔKE for the trip down.
iv) Calculate W_C, the work gravity did on her.
v) Work out ΔKE − W_C. Is it zero?
vi) Say what that number tells you. Name the force responsible and state how many joules it took.
vii) The slope is 60.0 m long. How big was that force?
viii) In the cart problem one term was zero and you went straight to the answer. Say in a sentence what the hill made you do that the bench did not.

When the force is not constant.

Everything above used W = Fd, which assumes the force held steady over the whole distance. When it does not — a spring, a hand that tires — the work is the area under the force-against-distance graph, $W = \int F \cdot dx$, taken along the path. A constant force is just the case where that area is a rectangle, and the

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integral collapses back to Fd. Note the variable of integration: work is kept over distance. Keep the same force over time instead and you get something different, which is where this Module goes next.



i) A force rises steadily from 0 N to 40 N over the first 5.0 m of a push. Sketch F against x and find the work from the area.

ii) Show that $\int F \cdot dx$ gives Fd when F is constant.

6) Looking ahead.

Work changes an object's energy, and it is a force acting over a distance. Later in this Module you will meet impulse, which changes an object's momentum, and which is a force acting over a time. Same force, same event — one set of books kept over distance, the other over time.

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